



NORTHEASTERN UNIVERSITY

IE 7374 Final Technical Report

August 2025

REFRIGERATOR FOOD TRACKER WITH RECIPE RECOMMENDER SYSTEM

SMART REFRIGERATOR AI PROJECT

GROUP 12

NORTHEASTERN UNIVERSITY

By Ahantya Vempati, Chantelle Chan, Jensen Ho, Shyam Patel

0



Table of Contents

ABSTRACT	2
1. INTRODUCTION	2
1.1 Motivation	2
1.2 Problem Statement	2
1.3 Objectives	3
2. DATASET	3
2.1 Dataset Overview	3
2.2 Filtering Criteria	4
2.3 Data Cleaning & Preprocessing	4
3. METHODOLOGY	5
3.1 Model Selection	5
3.2 Fine-Tuning Process	5
3.3 System Architecture	6
3.4 Implementation Details	6
4. EXPERIMENTAL SETUP	6
4.1 Hardware	6
4.2 Training Parameters	7
5. RESULTS	7
5.1 Quantitative Results	7
5.2 Qualitative Results	9
6. ANALYSIS	10
6.1 Strengths	10
6.2 Limitations	10
6.3 Error Analysis	10
7. CONCLUSION & FUTURE WORK	11
7.1 Conclusion	11
7.2 Future Work	11
REFERENCES	12

Abstract

Food waste is a global issue, so we decided to create a refrigerator system to handle this. We will be focused on NLP text-based GPT2 model for this project. This model was trained with the **RecipeNLG** dataset that we have chosen. There was the idea of having a vision-based model feeding data into a text-based model, but vision is beyond the scope of this project. The system takes ingredients and suggests recipes that the user can follow. Through this system the refrigerator can prioritize food for consumption before its expiry date. Our gpt2 model gpt2-recipes-manual_at_80pct was fine tuned with an 80 percent sub-sample of common refrigerator items such as fruit and vegetables from the RecipeNLG dataset. Using streamlit library, we built an interface to load the trained model for general use. Streamlit framework provides a developer server that allows quick testing of our application locally. Overall, we were quite satisfied with the result.

1. Introduction

1.1 Motivation

According to the United Nations Environment Program (2024), the world wastes over 1 billion meals' worth of edible food every day. To help address this issue, this project aims to implement a generative AI model designed to help households minimize food waste through intelligent inventory tracking, expiration alerts, and personalized meal recommendations.

Looking ahead, future innovations may involve the implementation of AI agent systems that extend beyond basic food tracking. These changes could make it possible for automated restocking, tailored nutrition monitoring, and even collaborating with composters that break down food waste at home. Ultimately, the refrigerator would evolve from a storage unit into a comprehensive, proactive tool for sustainable food management.

1.2 Problem Statement

Households in common day struggle with managing food waste, managing meals, and maintaining proper nutrition with the food they currently have on hand. Customers usually forget what items they brought in the refrigerators which in turn causes ingredients to be expired and in the end a waste of money. Not only that but spending time and creating healthy meals based on your ingredients can take a lot of time and be overwhelming.

Our project aims to fix this by developing an intelligent kitchen assistant that integrates real-time inventory tracking, expiration tracking, and personalized recipe generation. We are using generative AI, Retrieval Augmented Generation, Model fine-tuning, and transformer architectures. Through all this the assistant will create user-specific natural recipes, and track data on expiration of goods.

We aim to use our solution to fix issues with household efficiency by reducing the waste of goods and decreasing the amount of time for meal planning. In a world where we have inflation, health awareness, and sustainability issues our tool will drive smarter and adaptive decisions with food.

1.3 Objectives

- Implement **fine-tuned GPT-2** for recipe generation.
- Integrate **Refrigerator Inventory** tracking for factual accuracy.
- Provide a **Streamlit-based UI** for interaction.

2. Dataset

2.1 Dataset Overview

RecipeNLG (cooking recipes dataset)

RecipeNLG (cooking recipes dataset). (2020, December 9). Kaggle.

<https://www.kaggle.com/datasets/paultimothymooney/recipeNLG>

The dataset contains 2,231,142 entries across 7 columns. Each entry includes a hash (#) in the recipe identifier, a title, a list of ingredients, a direction (cooking instructions), a source URL link, and named entity recognition (NER) for ingredients.

Title (string)	Name of the recipe
Ingredients (string)	List of ingredients
Directions (string)	Cooking instructions
Link (string)	Source URL
Source (string)	Original content source
NER (string)	Named entity recognition labels
# (int)	Identifier for the recipe

2.2 Filtering Criteria

Filtering to Common Fridge Items:

Compiled a predefined list of commonly available fridge ingredients, including dairy products, vegetables, and fruits (e.g., milk, cheese, butter, yogurt, egg, tomato, onion, spinach, cabbage, zucchini, carrot, garlic, cucumber, pepper, celery, apple, grapes, corn, mushrooms, etc.) Recipes from the RecipeNLG dataset were filtered to meet the following criteria:

- Ingredient Count Constraint: Contain between 2 and 10 total ingredients to keep the recipes simple.
- Fridge Item Requirement: Contain at least one common fridge item from the predefined fridge list.

```
fridge_items = {
    "milk", "cheese", "cream cheese", "butter", "yogurt", "sour cream",
    "egg", "tomato", "onion", "lettuce", "spinach", "cabbage", "zucchini", "carrot", "garlic",
    "cucumber", "pepper", "celery", "apple", "grapes", "corn", "mushroom"}

def has_fridge_item(ingredients):
    match_count = sum(item.lower() in fridge_items for item in
    ingredients)
    return 2 <= len(ingredients) <= 10 and match_count >= 1
```

The purpose of this filter is to simulate a cooking scenario where a user has access to limit but common set of refrigerated ingredients. After filtering, then removed incomplete or low-quality entries to ensure that only recipes with usable instructions and ingredients lists were retained.

2.3 Data Cleaning & Preprocessing

The dataset for fine-tuning the GPT-2 model on recipe generation tasks. The goal of these steps was to ensure data quality, standardize the input-output format for training.

- The dataset cleaning and preprocessing involved the following steps:
 - Data Cleaning: Conducted data cleaning to eliminate inaccurate or incomplete entries. Remove rows with missing ingredients/directions, special characters. Ensure the dataset contained only complete and relevant recipes.
 - Normalize Measurement Units: All measurements units were standardized to a consistent format.

- **Tokenize:** Tokenize ingredients, used the GPT-2 tokenizer to convert raw text into token ID that suitable for model input.
- **Prompt-Response Standardization:** Standardized the data into a prompt response format <INPUT_START> ingredient1 <NEXT_INPUT> ingredient2 <INPUT_END> (e.g., “Ingredients: egg, flour, sugar → Recipe: [recipe text]”) to align the model’s training.
- **Model Saving:** The fine-tuned model was saved as gpt2-recipes-manual, ensure integration with Transformer, saved model weights can be reloaded into the GPT2LMHeadModel for inference and continued fine-tuned.
- **Tools:** Pandas, NLTK, PyTorch, Transformer, TQDM, JSON

```
print(f"Original dataset size: {len(df):,} recipes")
print(f"After fridge filter: {len(df_subset):,} recipes")
print(f"Train set: {len(df_train):,} recipes")
print(f"Test set: {len(df_test):,} recipes")
```

<i>Original dataset size:</i>	<i>2,231,142 recipes</i>
<i>After fridge filter:</i>	<i>27,394 recipes</i>
<i>Train set:</i>	<i>21,915 recipes</i>
<i>Test set:</i>	<i>5,479 recipes</i>

3. Methodology

3.1 Model Selection

GPT-2 (decoder-only transformer) chosen for open-source fine-tuning flexibility, because of its capacity to produce language that is consistent and logically structured, which make it ideal for creating comprehensive recipe with its instructions. It is suitable for this project of its conditional generation scenario because of its autoregressive transformer, which predicts each token depending on the previous context. For this project, the standard parameter model provided a great balance between output quality and computing efficiency.

3.2 Fine-Tuning Process

Fine-tune GPT-2 on the domain-specific recipe dataset with the following process:

- **Tokenizer Adjustment:** Add padding token pad_token to the GPT-2 tokenizer, ensuring batch processing without shape mismatches.
- **Training Mode:** Set the model to training mode via model.train() to enable gradient usability.
- **Batch Preparation:** Provide tokenized batches contain input_ids and attention masks.

- Loss Computation: Compute the cross entropy of loss by passing the tokenized sequences as inputs and labels, enable the model to learn to predict each token in the recipe.
- Optimization: Use the AdamW optimizer with low learning rate 5e-5 for stable fine-tuning.
- Backpropagation: Perform `loss.backward()` to calculate gradients, update weights with `optimizer.step()`, and clear gradients after each batch with `optimizer.zero_grad()`.
- Training Schedule: Train for 15 epochs, at the same time tracking average epoch loss to monitor convergence.

3.3 System Architecture

Dataflow Training in Colab:

```
RecipeNLG →
RecipeDataset → GPT2Tokenizer → GPT2LMHeadModel → model.save_pretrained() → test_model() → calculate_perplexity()
```

Inference in Local Machine with fine-tuned model:

```
Streamlit → Load_model() → generate_recipe() → update_inventory_from_prompt()
```

3.4 Implementation Details

- Frontend: Streamlit.
- Database: Application.
- Model Deployment: Local inference.
 - User Input via Streamlit – User enters available ingredients.
 - Inventory Tracker – Tracks items and expiration.
 - GPT-2 Model – Generates new recipes based on user input.
 - User Feedback – Accept/reject logic used for removing used items from inventory

4. Experimental Setup

4.1 Hardware

The model was trained on Colab and can run locally on Windows or MacOS.

- Google Colab Pro (T4 and L4)
- Local GPU on MacOS (Apple Silicon)
- Local GPU on Windows (Nvidia RTX).

4.2 Training Parameters

- Batch size: 2
- Learning rate: 0.00001
- Epochs: 15

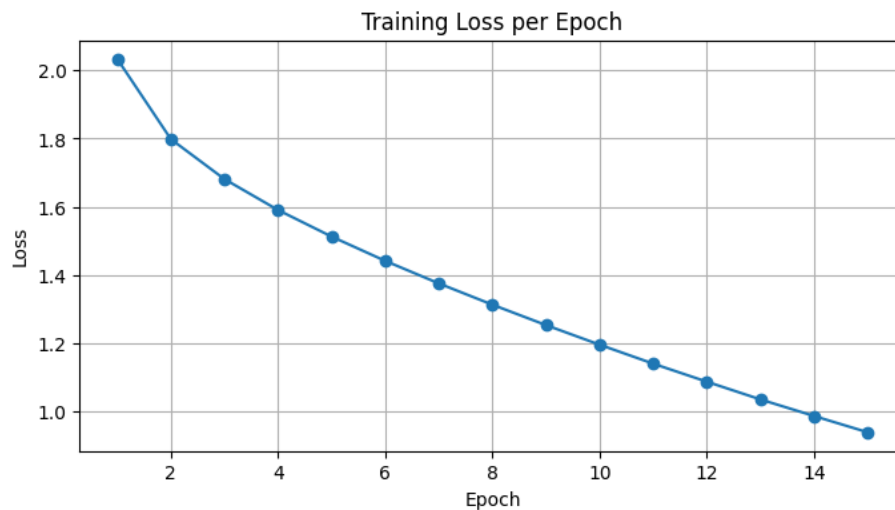


Figure 1 Training loss

5. Results

5.1 Quantitative Results

For the application, we measure the number of output that is acceptable then we apply user acceptance rate from recipe feedback.

Input	Output	Acceptable
egg, milk, tomato	Pizza Instructions: First you put some pieces of tomatoes on the plate and then you put a little bit (and bits) in your mouth and eat it."	Yes
egg, flour, sugar	American Slab , a Flat Egg Instructions: Mix flat egg with flour in a bowl and then add the sugar."	Yes
butter, garlic, mushroom	Broiled Mushrooms Instructions: Cut stems off mushrooms and place on broiler pan. Drizzle with olive oil, add chopped garlic and mushroom. Broil until tender."	Yes
egg, cheese, tomato, lettuce	Pizza Instructions: First you put some sauce on the plate and then you put your salad in the dish and then you put your	Yes

	cheese over it and then you toss some tomatoes and lettuce on it and then you throw a little olive on it and let it cook for 30 minutes at 100°."	
apple, grapes, yogurt	Fruit Smoothie Instructions: Pour yogurt in a bowl and add fruit."	Yes
zucchini, pepper, cucumber	Canned Zucchini Instructions: Cut zucchini in half lengthwise and remove seeds (use peeler to cut the bottom of each half). Place in a glass jar with 1/2 cup of water, cover and seal zucchini. Leave in zukes for 30 minutes. In a pan over medium heat add 3 tablespoons olive oil and sautee cucumbers until soft, about 5 minutes. Add pepper and salt to taste. Pour into serving bowl and garnish with a sprinkle Of Freshly Crushed Garamichoes	Yes
Corn, flour, tomato	Pizza Instructions: First you put the tomatoes and then the cucumber in the flour, then you put the cheese on it and finally let the dough rise until the dough is golden brown."	Yes
cream cheese, celery, tomato, egg	Pizza Cups Instructions: bake your pizza and then combine all the ingredients let cook for 5 minutes"	Yes
apple, cucumber, pepper, garlic	Cucumbers On The Grill Instructions: Peel cucumber and slice very thin rounds (1/2 inch thick) in a stick of cooking oil or other vegetable oil. Place slices on grill over medium heat until they begin to soften, then turn them over for about 20 minutes. Remove from the grill and let cool. Slice cucumber thinly. When peppers are soft enough that you can stick a fork into them easily when cooked with the stick; drain. Serve chilled with dressing."	Yes
butter, mushroom, carrot	Crab Cakes Instructions: 1) chop carrots and add them to a skillet 2-3T butter 3beat until soft 4 - 5Melt about 10 minutes til smooth Stir once or twice"	Yes

5.2 Qualitative Results

The Figure 2 below is example of generated recipes from fridge contents.

localhost:8501

 **Recipe AI Assistant**

 **Fridge Inventory Tracker**

Select Item

Quantity

Expiry Date

 Add Item

pepper

2

2025/08/09

Added 2 x pepper (expires 2025-08-09) to fridge.

Current Fridge Contents

	item	quantity	expiry
0	egg	4	2025-08-08
1	pepper	2	2025-08-09

Enter ingredients below and click 'Generate Recipe'

Your ingredients:

e.g., egg, flour, sugar

Generate Recipe

Stuffed Eggs

Instructions: Separate the whites from egg yolks and set aside for 30 minutes (you can then use a fork to break the yolks). Beat white until it forms soft peaks. Beat in egg whites. Add sugar and spices. Beat until fluffy."

 Accept Recipe

 Reject Recipe

 **Try these examples:**

 egg, flour, sugar

 butter, garlic, mushroom

 cheese, tomato, lettuce

 apple, grapes, yogurt

Figure 2 Application front-end with inventory tracking and recipe generation

6. Analysis

6.1 Strengths

Our system function quite well with streamlit and some guard railing (or output filtering). Generates creative, coherent recipes from minimal ingredients. The application works offline with no API dependency.

6.2 Limitations

Occasionally hallucinates unavailable ingredients or invalid recipes. We were able to mitigate this short coming with output filtering. Struggles with very sparse input. Overall, this architecture is limited by the older gpt2 model due to new openai-oss now available.

6.3 Error Analysis

During testing we found issue with recipe generator not returning valid recipes but instead we got food related URLs. To fix this we have a post generation function that check for suspicious result to guardrail against this type of behaviour. This is_suspicious function that can be expanded to check for future invalid input:

```
1. def is_suspicious(text):
2.     return re.search(r'https?:/', text, re.IGNORECASE) is not None
3. Inside the recipe_generator() function we retry up to 3 times for valid result
4. if not is_suspicious(result):
5.     return result
6. else:
7.     st.warning(f"Suspicious content detected. Retrying... (Attempt {attempt}/{max_retries})")
```

The reason this works is because retrying with the same prompt the model will generate a different output.

7. Conclusion & Future Work

7.1 Conclusion

Successfully developed open-source GPT-2 based recipe recommender with usable front-end using streamlit. Model fine-tuning and guard railing was demonstrated. Demonstrated sustainability benefits and usability in home kitchens.

7.2 Future Work

It is possible to integrate **vision-based fridge scanning** in the future. Improve nutritional analysis accuracy with Langchain Agentic AI combined with inventory analysis. Finally, deploy mobile app version that can connect with the fridge.

References

United Nations Environment Programme. (2024, March 27). *World squanders over 1 billion meals a day: UN report*. <https://www.unep.org/news-and-stories/press-release/world-squanders-over-1-billion-meals-day-un-report>

Basit, Abdul. (2024, December 7). Smart Storage: How Technology is Revolutionizing Food Preservation. <https://techbullion.com/smart-storage-how-technology-is-revolutionizing-food-preservation/>

Plessers. (n.d.). *How smart refrigerators are changing food storage*. https://www.plessers.com/blog/how_smart_refrigerators_are_changing_food_storage

RecipeNLG (cooking recipes dataset). (2020, December 9). Kaggle. <https://www.kaggle.com/datasets/paultimothymooney/recipeNLG>